Chinese Mandarin Speech Recognition Corpus

[AISHELL-ASR0009-OS1]

Training and Test Data

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1 Product Instruction

This Mandarin Chinese speech database [AISHELL-ASR0009] is main database. The utterance deals with 11 fields, such as smart home, unmanned driving and industrial production. The speakers from different accent areas in China are invited to participate in the recording. The process of recording in quiet indoor environment, using 3 different devices at the same time: high fidelity microphone (44.1kHz, 16bit); Android-system mobile phone (16kHz, 16bit), iOS-system mobile phone (16kHz, 16bit).

The manual transcription accuracy is above 95%, through professional speech annotation and strict quality inspection. We choose high fidelity microphone data from 400 people as as AISHELL-ASR0009-OS1 database and downsampled to 16kHz. The corpus contains 178 hours of speech, and is devided into training, developement and testing sets.

The data is free for academic use. Commercial use is forbidden.

2 Recording Text

2.1 Text Pool

2.1.1 Text Pool Content

Considering the application of speech recognition in smart home, unmanned driving, industrial production and other fields, the corpus is selected in 11 fields. (Chart 2-1)

Serial 6 to serial 10 are included in AISHELL-ASR0009-OS1.

Serial Number	**Field**
1	Smart Home Voice Control
2	POI (Geographic Information)
3	Music (Voice Control)
4	Digital String (Voice Control)
5	TV Play and Film Names
6	Finance
7	Science and Technology
8	Sports
9	Entertainments
10	News
11	English Spelling

Char 2-1 Text Pool Content

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2.1.2 Text Pool Processing

- Desensitization treatment. Delete politically sensitive, personal privacy, pornography, violence and other content.
- Delete inerpunctions such as < , > , [,] , ～ , / , \ , =, etc.
- Delete content in languages other than Chinese and English.
- Delete the sentence contains 25 words or more content.
- Unified format.

2.2 Structure Design of Recording Text

In view of speech coverage and phoneme balance, the recording text of this database is designed by the allocation of 500 sentences, extracted from the text pool, and structured as follows. The AISHELL-ASR0009-OS1 extraction of serial number 6 to 10. (Chart 2-2)

Serial Number	**Field**	**Allocation /#Sentence**
1	Smart Home Voice Control	5
2	POI (Geographic Information)	30
3	Music (Voice Control)	46
4	Digital String (Voice Control)	29
5	TV Play and Film Names	10
6	Finance	132
7	Science and Technology	85
8	Sports	66
9	Entertainments	27
10	News	66
11	English Spelling	4
Total	**11 fields**	**500 sentences**

Chart 2-2

3 Speaker Information

3.1 Speaker Information Registration

Speaker information include TaskID, Gender, Accent Area, Age Range and Birth Place. (Chart 3-1)

TaskID	**Gender**	**Accent Area**	**Age Range**	**Birth Place**
0001	M	North	A	Hebei

Chart 3-1

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TaskID: Each speaker can only fulfills 1 Task, while each Task corresponding to 1 recording text.

Gender: M defined as male, while F defined as female.

Accent Area: Divided into North, South, Yue-Gui-Min, and other areas according to the region where speakers belong to the native language.

Age range: A (16-25 years), B (26-40 years old), C (over 41 years old).

Birth Place: Duplicate from every speaker's Citizen ID Card.

3.2 Speaker Demographic Information

3.2.1 Gender Balance

This database consists of 186 male speakers and 214 female speakers.(Char 3-2-1)

Gender	Male	Female	Total
Percentage	47%	53%	**100%**

Chart 3-2-1

3.2.2 Age Distribution

A (16-25 years old) 316 people; B (26-40 years old) 71 people; C (41 years old or above) 13 people. (Chart 3-2-2)

	Age Range	#Speaker	Percentage	Male	Female
A	**16-25 yrs**	316	79%	140	176
B	**26-40 yrs**	71	18%	36	35
C	**> 41 yrs**	13	3%	10	3
Total		**400**	**100%**	**186**	**214**

Chart 3-2-2

3.2.3 Dialectal Regions

336 people in the North, 38 in the South, 18 in Yue-Gui-Min Area (Guangdong, Guizhou and Fujian province), 11 in other areas. (Chart 3-2-3)

Accent Area	#Speaker	%Speaker
North	333	83%
South	38	10%
Yue-Gui-Min	18	4%
Other Areas	11	3%

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合计	**400**	**100%**

Chart 3-2-3

4 Recording Processing

4.1 Recording Environment

Quiet indoors, not including other people voice, and other noises without reverberation. The speaker reads recording text at regular rate.

4.2 Recording Equipment

Recording equipment includes high fidelity microphone and recorder, iOS-system mobile phone, and Android-systems mobile phone.

4.3 Recording Method

The speaker is 20cm from the high-fidelity microphone and reads the recording text with the normal volume of normal speed. Android-system and iOS-system mobile phones respectively with microphone interval layout. (Chart 4-3)

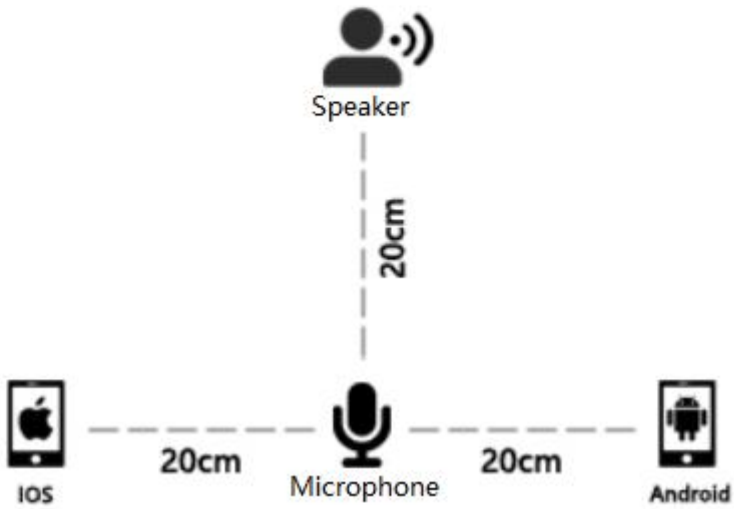

Chart 4-3

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5 Speech Content Annotation

Data annotator listen to the audio content to write, in order to make the text and audio content consistent pronunciation. General guidelines are as follows:

1) Transliteration and heard speech content must be completely consistent, not more, fewer, or wrongly written a word.

2) To transfer into digital form Chinese characters, such as "一二三", instead of "123". Pay attention to distinguish between "一" and "幺", "二" and "两".

3) Audio in English pronunciation should be written in the corresponding Chinese characters or English. Specific is divided into the following situations:

All the letters or words contained in the URL are capitalized. For example: the pronunciation content for the "www.abc.com", should transfer to "三 W 点 A B C 点 com"

The English pronunciation contains all lowercase words, transliteration.

English pronunce as words should transcript as lowercase.

English pronunce as spelling should transcript as uppercase.

For some proper nouns, or some English abbreviations, all transcript as uppercase with a space mark, such as C E O, C C T V, etc..

4) The integrity of the content should be consistent with the actual pronunciation, and shall not be deleted.

6 Corpus Catalog

6.1 Directory Structure

Directory tree	
Corpus Catalog	
│ AISHELL-ASR0009-OS1.docx	**(Text instruction)**
└─DOC	**(Description file)**
| | ├─all_wav_list.txt	**(Audio list)**
| | ├─content.txt	**(Transcripted content)**
| | ├─readme.txt	**(Catalog instruction)**
| | ├─spkrinfo.xlsx	**(Speaker information)**
|	
└─SPEECHDATA	
| ├─C0001	
|| || BAC0009C0001W0001.wav	
|| || BAC0009C0001W0001.txt	

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6.2 Naming Rule

6.2.1 Directory Naming Rules (Chart 6-2-1)

/<CORPUS>/<USAGE>/<FILE_ID>/<SPEECH_ID>

e.g. ChineseMandarinSpeechRecognitionCorpus/SPEECHDATA/C0001/BAC0009C0001W0001.wav

Directory	Content	Note
CORPUS	Chinese Mandarin Speech Recognition Corpus	Database Name
USAGE	SPEECHDATA	Folder Name
FILE_ID	C0001	SpeakerID Folder
SENTENCE_ID	BAC0009C0001W0001.txt	TXT documents
SPEECH_ID	BAC0009C0001W0001.wav	WAV documents

Chart 6-2-1

6.2.2 File Naming Rules (Chart 6-2-2)

<CORPUS_ID><SPEAKER_IC><WAV_NUM>

e.g. BAC0009C0001W0001.wav

File	Content	Note
CORPUS_ID	BAC009	Database min
SPEAKER_NUM	C0001	Speaker ID
WAV_NUM	W0001	WAV number

Chart 6-2-2

7 AISHELL-ASR0009-OS1 Dataset

This experiment extracted five categories of financial, technical, sports, entertainment and current news items of 400 people, and carried out the baseline evaluation of speech recognition system.

Allocate concretely:

Subset	Hours	Male speakers	Female speakers	Total speakers
Training	150	161	179	340
Development	18	12	28	40
Test	10	13	7	20

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